Undergraduate Medicine

Cardiology Study Pack

CharaNas Medicine — Specialty Notes

Expanded notes with original schematic figures for revision. These are educational drawings inspired by standard undergraduate teaching themes, not copied textbook plates. Always follow your faculty curriculum and latest guidelines.

Section	Focus
1	Core physiology & anatomy
2	Ischemic heart disease & ECG ideas
3	Heart failure & murmurs
4	Arrhythmias & emergencies
5	Ward checklist & book anchors
6	Quick pearls from these notes
7	Recommended book sources

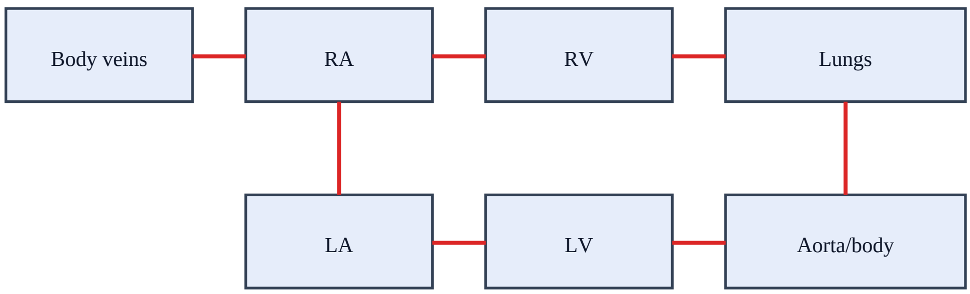
1. Core physiology & anatomy

Cardiac output (CO) = heart rate × stroke volume. Mean arterial pressure approximates CO × systemic vascular resistance. The Frank–Starling mechanism links increased venous return to increased stroke volume within physiologic limits.

Coronary perfusion of the left ventricle occurs mainly in diastole. The right coronary artery usually supplies the SA node. Know the territories: LAD → anterior wall; RCA → inferior wall; circumflex → lateral wall.

Valves: mitral and tricuspid are atrioventricular; aortic and pulmonic are semilunar. Auscultation points and radiation patterns are high-yield for undergraduate exams and wards.

Figure: Simplified cardiac blood-flow pathway



Deoxygenated path above → lungs; oxygenated path below → systemic circulation

Schematic figure for learning — verify details in your recommended textbooks.

Revision prompts

- Explain this section aloud in 60 seconds as if teaching a junior student about cardiology.
- List 3 red flags that would make you escalate care urgently.
- Name 2 investigations and 2 initial management steps you would consider first.

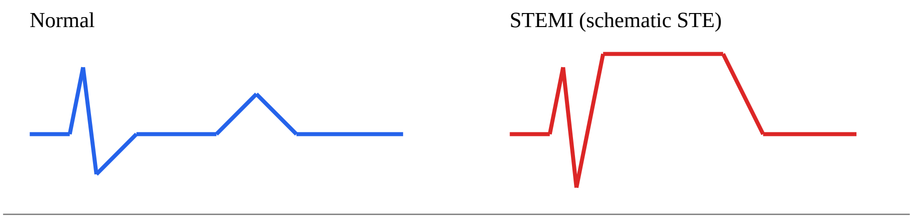
2. Ischemic heart disease & ECG ideas

Chronic coronary disease presents with exertional angina; acute coronary syndromes include unstable angina, NSTEMI, and STEMI. Time-critical reperfusion saves myocardium in STEMI.

STEMI: ST elevation in contiguous leads (e.g., II/III/aVF inferior; V2–V4 anterior). NSTEMI: troponin rise without STEMI criteria. Always combine ECG with history, exam, and risk factors.

Immediate priorities in ACS: ABCs, ECG within minutes, aspirin (if no contraindication), anticoagulation/antiplatelet strategy per protocol, and urgent reperfusion for STEMI.

Figure: Schematic STEMI vs normal ST segment



Educational schematic only — interpret real ECGs with clinical context

Schematic figure for learning — verify details in your recommended textbooks.

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3. Heart failure & murmurs

Heart failure symptoms: dyspnea, orthopnea, PND, edema. Signs: raised JVP, crackles, S3, peripheral edema. BNP/NT-proBNP and echocardiography help confirm and classify HFrEF vs HFpEF.

HFrEF cornerstone therapy commonly includes ACEi/ARB/ARNI, evidence-based beta-blocker, mineralocorticoid receptor antagonist, and SGLT2 inhibitor, with loop diuretics for congestion.

Mitral regurgitation: holosystolic murmur to axilla. Aortic stenosis: crescendo–decrescendo systolic murmur to carotids, slow-rising pulse. Aortic regurgitation: early diastolic decrescendo. Mitral stenosis: diastolic rumble (± opening snap).

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4. Arrhythmias & emergencies

AF: irregularly irregular pulse; risk-stratify stroke risk (e.g., CHA2DS2-VASc) and bleeding risk when considering anticoagulation. Rate vs rhythm control depends on context.

Unstable tachycardia/bradycardia: follow ACLS-style pathways — if unstable with serious signs, prepare for synchronized cardioversion or pacing as indicated.

Pericarditis: positional/pleuritic pain, diffuse STE with PR depression. Tamponade: Beck triad (hypotension, raised JVP, muffled sounds) — urgent echo and drainage.

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5. Ward checklist & book anchors

Ward checklist: vitals + SpO2, focused CV exam, ECG, troponin if ACS suspected, CXR if HF/infection, echo when structural disease/HF suspected, medication review (beta-blockers, ACEi, diuretics, anticoagulants).

Secondary prevention after MI: lifestyle, high-intensity statin when appropriate, antiplatelet therapy, BP/glucose control, cardiac rehab.

Book anchors: Braunwald's Heart Disease; Harrison's cardiology sections; ECG Made Easy (Hampton). Prefer faculty-recommended editions.

Revision prompts

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6. Quick pearls

High-yield reminders packaged with this specialty in CharaNas:

1. ACS spectrum: unstable angina, NSTEMI, STEMI — treat as time-critical.

2. STEMI: ST elevation in contiguous leads ± new LBBB; urgent reperfusion.

3. HF signs: orthopnea, raised JVP, crackles, edema; confirm with BNP/echo.

4. MR = holosystolic to axilla; AS = crescendo-decrescendo to carotids.

5. Secondary prevention after MI: antiplatelet, statin, beta-blocker, ACEi as indicated.

- Link symptoms → anatomy/physiology → investigation → first management step.
- Write one OSCE-style explanation for a common presentation in this specialty.
- After reading, do the Short MCQ and Case-based Question for active recall.

7. Recommended book sources

Standard undergraduate references commonly used internationally. Use the edition your college recommends. Figures in commercial textbooks are copyrighted; use library/licensed access for full plates and photographs.

#	Reference
1	Braunwald's Heart Disease
2	Harrison's Principles of Internal Medicine — Cardiology
3	ECG Made Easy — John Hampton

How to study with this PDF: read one section → sketch the figure from memory → answer the MCQs/cases → review mistakes → revisit the matching section.